

Independent Study

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This Week

- Finished Setting up FAISS RAG in LTRAG
- Journals and Publishing

FAISS RAG

The original codebase used FastGPT for RAG. I switched it to FAISS RAG. I used aspects from my modified LINC, where I used FAISS RAG. We create a FAISS vector store, use OpenAI embeddings to embed, and then retrieve k relevant examples to use for our few-shot prompts. This code primarily lives in `utils/knowledge.py` for both frameworks (FOLIO and LSAT). Please refer to myLTRAG in [aiea-rami](#).

FastGPT vs. FAISS

A minor limitation here is FAISS and FastGPT use different similarity searches and build vector stores differently. This gap might be minimal because they still function similarly. However, this might be one of the reasons we aren't able to recreate LTRAG's results perfectly since RAG is a huge part of this paper. Just something to keep in mind. Also, LTRAG never mentions the embedding model used, so I am resorting to using a popular one at the time of writing their paper (ada-002).

Journals and Publishing

I wrote a [small paper](#) on AI bias about a year ago. I was looking to use it as a way to better understand the publishing world and how papers work. This journey has been very meaningful to me. I learned what SSRN is and how it differs from ARXIV. I learned about popular journals and conferences like JAIR and NEURIPS.

To get a start, I decided to create an SSRN account and upload my small paper on it just to test the waters. I also started looking at appropriate journals for it. I have come to realize that literature review-style papers are not appropriate for journals like JAIR, which seem to publish more original research, looking at their website. I am now looking for smaller journals such as AJUR. Hopefully I'll find something good soon. I just want to try this out before dealing with a paper of higher stakes like this project.

Conclusion/Future Work

This week I learned a lot about research. I incorporated elements from my previous work onto LTRAG. I also learned a lot about publishing and different journals. As for next week, I will have to run larger-scale tests to make sure that all methods (LTRAG, COT, and standard) work in both frameworks (FOLIO and LSAT) and do more reading.

References

AI Use

I used Claude for:

1. Explaining pre-existing code. For example, understanding FastGPT and how it relates to FAISS RAG.
2. Explaining how to use certain functions: It helped me understand how FastGPT is used and all the appropriate functions and how they relate to FAISS RAG's from LangChain.
3. Debugging: Fixing errors related to setting up FAISS RAG.

All changes were made by the author.